CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M.Tech.)
SEM - I University Examination, May- 2011
M.Tech. (CAD/CAM)
ME – 703 Advanced Dynamics of Machines

Date: 11.05.2011, Wednesday **Time: 10:00 a.m. to01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. Figures to the right indicate *full* marks.
2. Make suitable assumptions and draw neat figures wherever if required.

SECTION- I

Q1 (a) A three cylinder radial engine has cylinders located 120^0 from neighboring cylinder. Reciprocating mass of each cylinder is 1.2kg. Length of crank is 75mm and each connecting rod is 250mm long. Find out maximum primary and secondary unbalance forces if the engine runs at 25000 r.p.m. **10**

(b) Explain the Active balancing and passive balancing. **02**

OR

Q1 (a) The firing order of a 2-stroke cycle six cylinder diesel engine is I-V-III-VI-II-IV. The Adjacent cylinder centerlines are 375 mm apart. Each cylinder has a connecting rod 600mm long and a stroke 225mm. The mass of the reciprocating parts in each cylinder is 100 kg. Determine the magnitudes of the primary and secondary unbalanced forces and moments when the engine runs at a constant speed of 250 rpm. **12**

Q2 (a) ABCD is a four bar mechanism. Link length AB=50mm, BC=90mm, CD=110mm,Fixed link DA=125mm and $\angle BAD = 50^0$. If the link AB rotates at constant angular velocity of 12 rad/sec. Determine velocity and acceleration of other links by vector and space method. Compare the results. **12**

OR

Q2 (a) Fig.1 shows an offset slider crank mechanism. Determine displacement, velocity and acceleration of coupler point P and slider B for $\theta = 50^0$, OA=50, AB=120, e=20, AP=BP, $\angle APB = 90^0$ assume $\omega_a = 1 rad/\sec$. Use analytical method. **12**

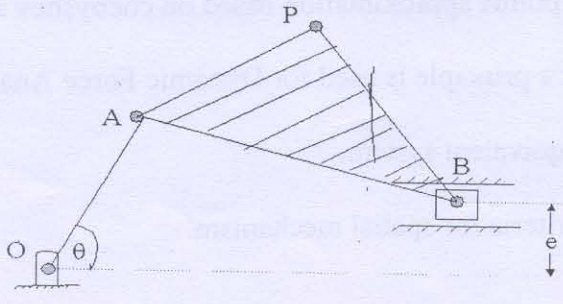

Fig.1

Q3 (a) In 3-4-5 Polynomial cam follower rise 20mm in 120° of cam rotation. The cam shaft rotates at 300 rpm. Find Displacement, Velocity, Acceleration and Jerk when shaft has turned through 35^0. **06**

P.T.O

(b) Explain the jump Phenomenon. How will you determine jump speed for an eccentric cam operating a flat faced follower? **05**

SECTION - II

Q4 Fig 2 Shows a Mechanism ABCD, EFG are the midpoints of the links as shown in fig. Determine the torque T2 to keep the mechanism in static equilibrium. Also find pin reaction (By using Analytical Method). AB = 30 mm, BC = 90 mm, CD = 60 mm, AD = 40 mm. **12**

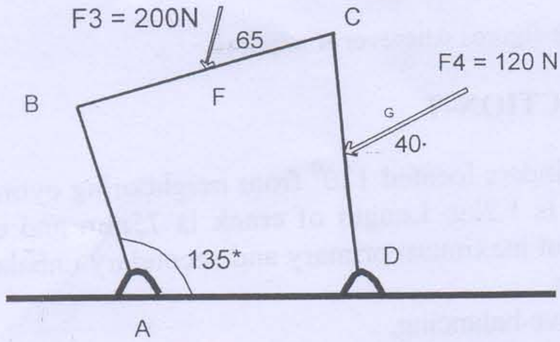

Fig.2

OR

Q4 (a) Explain the consideration of sliding friction in static force analysis with suitable example. **02**

Q4 (b) In Quick return mechanism shown in figure 1. the force 1 KN acting perpendicular to DB at its mid point in inward direction. Determine the static equilibrium torque on link OA by virtual work method. **10**

Q5 Synthesis a four bar mechanism for the following position of input and output link. **12**

$$\theta = 90^0, \phi = 60^0$$

$$\omega_2 = 2rad/\sec, \omega_4^{\cdot} = 3rad/\sec$$

$$\alpha_3 = 0rad/\sec^2, \alpha_4 = -1rad/\sec^2$$

OR

Q5 A 4-bar linkage is required to generate a function $y=x^{1.6}$ for $1 \le x \le 3$. The crank rotates from an angle 60^0 to 120^0; where as the follower rotates from an angle 60^0 to 150^0. Use three accuracy points approximation based on chebyshev spacing. **12**

Q6 (a) Explain how D'Alembert's principle is used for Dynamic Force Analysis **03**

(b) Explain the Dynamical Equivalent system. **03**

(c) Explain the Kutzbach's criteria for spatial mechanism. **05**